

PROBLEM SET 2—SOLUTIONS

ECO403/503—INTRODUCTION TO CONTRACT THEORY AND AUCTIONS

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Instructions: These are my suggested solutions. It is your responsibility to check this document against your own solutions.

1. (a) If α^* is an optimum, it must satisfy the first order condition:

$$-q(1-\pi)u'(w-\alpha^*q) + \pi(1-q)u'(w-D+\alpha^*(1-q)) \leq 0$$

with equality if $\alpha^* > 0$. The above equation is equivalent to

$$\frac{u'(w-D+\alpha^*(1-q))}{u'(w-\alpha^*q)} \leq \frac{q(1-\pi)}{\pi(1-q)}$$

with equality if $\alpha^* > 0$.

Remark: since $q > \pi$ this implies that $1-q < 1-\pi$ which implies that $1 < \min\left\{\frac{q}{\pi}, \frac{1-\pi}{1-q}\right\}$ which implies that $1 < \frac{q(1-\pi)}{\pi(1-q)}$.

We will now consider two cases:

Case 1: $\alpha^* = 0$. If this is the case, we have from the first order condition that

$$1 < \frac{u'(w-D)}{u'(w)} \leq \frac{q(1-\pi)}{\pi(1-q)}$$

where the first inequality is due to the fact that u is strictly increasing and concave. Therefore, we conclude that if $\frac{u'(w-D)}{u'(w)} > \frac{q(1-\pi)}{\pi(1-q)}$, $\alpha^* = 0$ cannot be a solution, but if $\frac{u'(w-D)}{u'(w)} \leq \frac{q(1-\pi)}{\pi(1-q)}$, then $\alpha^* = 0$ is a solution. Thus, we conclude that $\alpha^* = 0 < D$.

Case 2: $\alpha^* > 0$. If this is the case, then the first order condition is satisfied with equality:

$$\frac{u'(w-D+\alpha^*(1-q))}{u'(w-\alpha^*q)} = \frac{q(1-\pi)}{\pi(1-q)} > 1$$

where the last inequality comes from the remark above. using this fact, we get that

$$u'(w-D+\alpha^*(1-q)) > u'(w-\alpha^*q) \Rightarrow w-D+\alpha^*(1-q) < w-\alpha^*q \Rightarrow \alpha^* < D,$$

where the above implications follow from u being strictly increasing and concave.

- (b) *Comparative Statics.* Suppose that $\alpha^* > 0$, so we will take that the first order condition is like in case 2 above, i.e., it holds with equality at the optimum α^* :

$$\frac{u'(w-D+\alpha^*(1-q))}{u'(w-\alpha^*q)} = \frac{q(1-\pi)}{\pi(1-q)} \tag{1}$$

Define $F : \mathbb{R}_+ \times \mathbb{R}_+^2 \times [0, 1]^2 \rightarrow \mathbb{R}$ as $F(\alpha, W, D, q, \pi) = \pi(1-q)u'(w-D+\alpha(1-q)) - q(1-\pi)u'(w-\alpha q)$ and rewrite equation (1) as:

$$F(\alpha^*, w, D, q, \pi) = \pi(1-q)u'(w-D+\alpha^*(1-q)) - q(1-\pi)u'(w-\alpha^*q) = 0.$$

We can use the implicit function theorem to calculate the derivatives

$$\frac{\partial \alpha^*(w, D, q, \pi)}{\partial w} \quad \text{and} \quad \frac{\partial \alpha^*(w, D, q, \pi)}{\partial q},$$

and in order to do this, we will need the following partial derivatives of F : $F_\alpha(\alpha, w, D, q, \pi)$, $F_w(\alpha, w, D, q, \pi)$ and $F_q(\alpha, w, D, q, \pi)$:

$$F_\alpha(\alpha, w, D, q, \pi) = \pi(1-q)^2 u''(w - D + \alpha(1-q)) + q^2(1-\pi) u''(w - \alpha q) \quad (2)$$

$$F_w(\alpha, w, D, q, \pi) = \pi(1-q) u''(w - D + \alpha(1-q)) - q(1-\pi) u''(w - \alpha q) \quad (3)$$

$$F_q(\alpha, w, D, q, \pi) = -\alpha [\pi(1-q) u''(w - D + \alpha(1-q)) - q(1-\pi) u''(w - \alpha q)] - [\pi u'(w - D + \alpha(1-q)) + (1-\pi) u'(w - \alpha q)] \quad (4)$$

In order to calculate the signs of $\partial \alpha^* / \partial w$ and $\partial \alpha^* / \partial q$, we will need to put a sign on (2), (3) and (4). This is not possible for general values of α, W, D, q, π , but since we only need the sign at the optimal value α^* , we can do this by using the first-order condition (1).

- $F_\alpha(\alpha^*, w, D, q) < 0$ because it is a sum of strictly negative terms: since u is strictly concave, then $u''(x) < 0$ for all x .
- To put a sign on $F_w(\alpha^*, w, d, q) = \pi(1-q) u''(w - D + \alpha^*(1-q)) - q(1-\pi) u''(w - \alpha^* q) < 0$, let us first see what would it require to be negative:

$$\begin{aligned} & \pi(1-q) u''(w - D + \alpha^*(1-q)) - q(1-\pi) u''(w - \alpha^* q) < 0 \\ \Leftrightarrow & \pi(1-q) u''(w - D + \alpha^*(1-q)) < q(1-\pi) u''(w - \alpha^* q) \\ \Leftrightarrow & \frac{u''(w - D + \alpha^*(1-q))}{u''(w - \alpha^* q)} > \frac{q(1-\pi)}{\pi(1-q)} \end{aligned}$$

where the change in the direction of the inequality comes from u'' being strictly negative,

$$\Leftrightarrow \frac{u''(w - D + \alpha^*(1-q))}{u''(w - \alpha^* q)} > \frac{u'(w - D + \alpha^*(1-q))}{u'(w - \alpha^* q)}$$

where the right-hand side substitution comes from using the first-order condition (1)

$$\Leftrightarrow \frac{u''(w - D + \alpha^*(1-q))}{u'(w - D + \alpha^*(1-q))} < \frac{u''(w - \alpha^* q)}{u'(w - \alpha^* q)}$$

where the change in the direction of the inequality again comes from u'' being strictly negative,

$$\Leftrightarrow \frac{u''(a)}{u'(a)} < \frac{u''(b)}{u'(b)}$$

where we have made the substitution $a = w - D + \alpha^*(1-q)$ and $b = w - \alpha^* q$,

$$\begin{aligned} \Leftrightarrow & -\frac{u''(a)}{u'(a)} > -\frac{u''(b)}{u'(b)} \\ \Leftrightarrow & r_A(a, u) > r_A(b, u) \end{aligned}$$

where the last line comes from the definition of the *coefficient of absolute risk aversion*. Since, from part (a), we know that $a < b$, then $F_w(\alpha^*, w, D, q) < 0$ if and only if the utility function u exhibits *Decreasing Absolute Risk Aversion (DARA)*. Similarly, $F_w(\alpha^*, w, D, q) > 0$ if and only if u exhibits *Increasing Absolute Risk Aversion (IARA)*, and finally $F_w(\alpha^*, w, D, q) = 0$ if and only if u exhibits *Constant Relative Risk Aversion (CARA)*.

- $F_q(\alpha^*, w, D, q, \pi)$ is given by

$$\begin{aligned} & -\alpha^* [\pi(1-q) u''(w - D + \alpha^*(1-q)) - q(1-\pi) u''(w - \alpha^* q)] \\ & - [\pi u'(w - D + \alpha^*(1-q)) + (1-\pi) u'(w - \alpha^* q)]. \end{aligned}$$

Putting a sign to this expression turns out to be tricky, so we will analyze each term on its own and then collect the results:

- i. For the component $-\alpha^*$, note that since $\alpha^* > D > 0$, this term is negative.
- ii. For the component $[\pi(1-q)u''(w-D+\alpha^*(1-q)) - q(1-\pi)u''(w-\alpha^*q)]$ we have seen, from our work in putting a sign to $F_w(\alpha^*, w, D, q, \pi)$, that
 - $[\pi(1-q)u''(w-D+\alpha^*(1-q)) - q(1-\pi)u''(w-\alpha^*q)] < 0$ if and only if u is DARA.
 - $[\pi(1-q)u''(w-D+\alpha^*(1-q)) - q(1-\pi)u''(w-\alpha^*q)] = 0$ if and only if u is CARA.
 - $[\pi(1-q)u''(w-D+\alpha^*(1-q)) - q(1-\pi)u''(w-\alpha^*q)] > 0$ if and only if u is IARA.
- iii. For the component $-\pi u'(w-D+\alpha(1-q)) + (1-\pi)u'(w-\alpha q)$, note that the term inside the brackets is strictly positive since it is a sum of strictly positive terms, given that $u' > 0$ for all values and $\pi > 0$. Then the whole term, together with the negative sign outside the brackets, is negative.

Collecting the three results imply that:

- $F_q(\alpha^*, w, D, q, \pi) < 0$ if u is IARA or CARA, since it is a sum of only negative terms.
- If u is DARA, we cannot sign $F_q(\alpha^*, w, D, q, \pi)$ on the assumptions we have made until now. However, for standard DARA utility functions u , $F_q(\alpha^*, w, D, q, \pi)$ will be negative.

Collecting all the information above, and using the Implicit Function Theorem, we have that:

$$\frac{\partial \alpha^*}{\partial w} = -\frac{F_w(\alpha^*, w, D, q, \pi)}{F_\alpha(\alpha^*, w, D, q, \pi)} \begin{cases} < 0 & \text{if and only if } u \text{ is DARA} \\ = 0 & \text{if and only if } u \text{ is CARA} \\ > 0 & \text{if and only if } u \text{ is IARA} \end{cases}$$

Note that α^* is then *inferior to wealth* if u is DARA, *neutral to wealth* if u is CARA, and *normal to wealth* if u is IARA.

On the other hand,

$$\frac{\partial \alpha^*}{\partial q} = -\frac{F_q(\alpha^*, w, D, q, \pi)}{F_\alpha(\alpha^*, w, D, q, \pi)} \begin{cases} < 0 & \text{if } u \text{ is CARA or IARA} \\ \text{undefined} & \text{if } u \text{ is DARA} \end{cases}$$

However, for most standard cases of *DARA* utility functions, $\partial \alpha^* / \partial q < 0$. Therefore, α^* decreases when the price of insurance q increases, so the demand for insurance follows the law of demand.

- (c) For each individual, their first-order conditions are (assuming interior solutions)

$$\phi_1(\alpha_1^*) = -q(1-\pi)u_1'(w-\alpha_1^*q) + \pi(1-q)u_1'(w-D+\alpha_1^*(1-q)) = 0 \quad (5)$$

$$\phi_1(\alpha_2^*) = -q(1-\pi)u_2'(w-\alpha_2^*q) + \pi(1-q)u_2'(w-D+\alpha_2^*(1-q)) = 0 \quad (6)$$

The concavity of u_1 implies that ϕ_1 is decreasing. Therefore, if we show that $\phi_1(\alpha_2^*) > 0$, it must follow that $\alpha_1^* > \alpha_2^*$. Since $u_1(x) = \Psi(u_2(x))$ for $\Psi : \mathbb{R} \rightarrow \mathbb{R}$ strictly increasing and strictly concave, we have that

$$\phi_1(\alpha_2^*) = -q(1-\pi)\Psi'(u_2(w-\alpha_2^*q))u_2'(w-\alpha_2^*q) + \pi(1-q)\Psi'(u_2(w-D+\alpha_2^*(1-q)))u_2'(w-D+\alpha_2^*(1-q))$$

Note that in the expression above, all the terms are the same as in $\phi_2(\alpha_2^*)$ but with different weights: the negative term $-q(1-\pi)u_2'(w-\alpha_2^*q)$ is weighted with $\Psi'(u_2(w-\alpha_2^*q))$; and the positive term $\pi(1-q)u_2'(w-D+\alpha_2^*(1-q))$ is weighted with $\Psi'(u_2(w-D+\alpha_2^*(1-q)))$. Therefore, $\phi_1(\alpha_2^*) > 0$ if the positive terms in $\phi_2(\alpha_2^*)$ are weighted more than the negative terms. This is equivalent to

$$\Psi'(u_2(w-\alpha_2^*q)) < \Psi'(u_2(w-D+\alpha_2^*(1-q))).$$

since Ψ' is strictly positive and decreasing (because Ψ is strictly increasing and concave). Thus, the previous expression is equivalent to

$$u_2(w-\alpha_2^*q) > u_2(w-D+\alpha_2^*(1-q)) \Leftrightarrow w-\alpha_2^*q > w-D+\alpha_2^*(1-q) \Leftrightarrow \alpha_2^* < D,$$

where the middle inequality is true since u_2 is a strictly increasing function.

Hence, we conclude that $\phi_1(\alpha_2^*) > 0$ which implies that $\alpha_1^* > \alpha_2^*$. Thus, individual 1, who is more risk-averse than 2, insures a larger amount.

2. (a) We have that $u'(x) = (1/2)x^{-1/2}$, and that $u''(x) = -(1/4)x^{-3/2}$. Thus, $r_A(x, u) = (1/2)x^{-1}$. Therefore, $r_A(5, u) = 1/10$.

(b) The expected value of this lottery is 10. For the certainty equivalent, we need to solve the equation

$$\sqrt{c_1} = (1/2)\sqrt{16} + (1/2)\sqrt{4} \Rightarrow c_1 = 9.$$

Thus $RP_1 = 10 - 9 = 1$.

(c) The expected value of this lottery is 26. For the certainty equivalent we need to solve the equation

$$\sqrt{c_2} = (1/2)\sqrt{36} + (1/2)\sqrt{16} \Rightarrow c_2 = 25.$$

Thus, $RP_2 = 26 - 25 = 1$.

3. (a) The certainty equivalent is the solution to

$$u(c(x, u, \epsilon)) = (1/2)u(x + \epsilon) + (1/2)u(x - \epsilon) \quad (7)$$

$$c(x, u, \epsilon) = u^{-1}((1/2)u(x + \epsilon) + (1/2)u(x - \epsilon)) \quad (8)$$

(b) To obtain the derivatives, it will be easier to use *implicit differentiation* over equation (7). Differentiating (7) with respect to ϵ yields:

$$u'(c(x, u, \epsilon)) \frac{\partial c(x, u, \epsilon)}{\partial \epsilon} = \frac{1}{2} [u'(x + \epsilon) - u'(x - \epsilon)], \quad (9)$$

$$\frac{\partial c(x, u, \epsilon)}{\partial \epsilon} = \frac{\frac{1}{2} [u'(x + \epsilon) - u'(x - \epsilon)]}{u'(c(x, u, \epsilon))}, \quad (10)$$

where in (10) we have used (8) to simplify it. Implicit differentiation of (9) yields:

$$u''(c(x, u, \epsilon)) \left(\frac{\partial c(x, u, \epsilon)}{\partial \epsilon} \right)^2 + \left(\frac{\partial^2 c(x, u, \epsilon)}{\partial \epsilon^2} \right) u'(c(x, u, \epsilon)) = \frac{1}{2} [u''(x + \epsilon) + u''(x - \epsilon)], \quad (11)$$

$$\frac{\partial^2 c(x, u, \epsilon)}{\partial \epsilon^2} = \frac{\frac{1}{2} [u''(x + \epsilon) + u''(x - \epsilon)] - u''(c(x, u, \epsilon)) \left(\frac{\partial c(x, u, \epsilon)}{\partial \epsilon} \right)^2}{u'(c(x, u, \epsilon))}. \quad (12)$$

Now, note that

$$\lim_{\epsilon \rightarrow 0} c(x, u, \epsilon) = u^{-1}((1/2)u(x) + (1/2)u(x)) = u^{-1}(u(x)) = x, \quad (13)$$

$$\lim_{\epsilon \rightarrow 0} \frac{\partial c(x, u, \epsilon)}{\partial \epsilon} = \frac{\frac{1}{2} [u'(x) - u'(x)]}{u'(x)} = 0. \quad (14)$$

Using (13) and (14) into (12) when taking the limit yields:

$$\lim_{\epsilon \rightarrow 0} \frac{\partial^2 c(x, u, \epsilon)}{\partial \epsilon^2} = \frac{\frac{1}{2} [u''(x) + u''(x)] - u''(x) \cdot 0}{u'(x)} = \frac{u''(x)}{u'(x)} = -r_A(x, u) \quad (15)$$

4. (a) The expected utility of the portfolio $\alpha z + \beta$, $E[u(\alpha z + \beta)]$ is given by:

$$\begin{aligned}
&= \int_{-\infty}^{\infty} -\exp[-\gamma(\alpha z + \beta)] \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{1}{2}\left(\frac{z-\mu}{\sigma}\right)^2\right] dz \\
&= \int_{-\infty}^{\infty} -\exp(-\gamma\beta) \exp\left\{-\gamma\alpha z + \left[-\frac{1}{2}\left(\frac{z-\mu}{\sigma}\right)^2\right]\right\} \frac{1}{\sqrt{2\pi\sigma^2}} dz \\
&= -\exp(-\gamma\beta) \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2}\left[\left(\frac{z-\mu}{\sigma}\right)^2 + 2\gamma\alpha z\right]\right\} dz \\
&= -\exp(-\gamma\beta) \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2}\left[\frac{z^2 - 2z\mu + \mu^2 + 2\gamma\sigma^2\alpha z}{\sigma^2}\right]\right\} dz \\
&= -\exp(-\gamma\beta) \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2}\left[\frac{z^2 - 2z(\mu - \gamma\sigma^2\alpha) + (\mu - \gamma\sigma^2\alpha)^2 + 2\mu\gamma\sigma^2\alpha - \gamma^2\sigma^4\alpha^2}{\sigma^2}\right]\right\} dz \\
&= -\exp(-\gamma\beta) \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2}\left[\frac{(z - (\mu - \gamma\sigma^2\alpha))^2}{\sigma^2} - \mu\gamma\alpha + \frac{\gamma^2\sigma^2\alpha^2}{2}\right]\right\} dz \\
&= -\exp(-\gamma\beta) \exp\left(-\mu\gamma\alpha + \frac{\gamma^2\sigma^2\alpha^2}{2}\right) \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2}\left[\frac{(z - (\mu - \gamma\sigma^2\alpha))^2}{\sigma^2}\right]\right\} dz \\
&= -\exp\left[-\gamma\left(\beta + \mu\alpha - \frac{\gamma\sigma^2\alpha^2}{2}\right)\right] \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2}\left[\frac{(z - (\mu - \gamma\sigma^2\alpha))^2}{\sigma^2}\right]\right\} dz \\
&= -\exp\left[-\gamma\left(\beta + \mu\alpha - \frac{\gamma\sigma^2\alpha^2}{2}\right)\right]
\end{aligned}$$

where the last equality is obtained because the integrand in the previous equation is a normal distribution with mean $\mu - \gamma\sigma^2\alpha$ and variance σ^2 and therefore, integrates to 1.

Therefore, the maximization problem of the agent becomes

$$\begin{aligned}
&\max_{\alpha, \beta \geq 0} -\exp\left[-\gamma\left(\beta + \mu\alpha - \frac{\gamma\sigma^2\alpha^2}{2}\right)\right] \\
&\text{subject to } \alpha + \beta \leq w.
\end{aligned}$$

We can transform the objective function with the monotonic transformation $f(x) = -\ln(-x)/\gamma$ to yield the following maximization problem:

$$\begin{aligned}
&\max_{\alpha, \beta \geq 0} \beta + \mu\alpha - \frac{\gamma\sigma^2\alpha^2}{2} \\
&\text{subject to } \alpha + \beta \leq w.
\end{aligned}$$

As in the lecture, we can write the problem above as

$$\begin{aligned}
&\max_{\alpha} w + \alpha(\mu - 1) - \frac{\gamma\sigma^2\alpha^2}{2} \\
&\text{subject to } 0 \leq \alpha \leq w.
\end{aligned}$$

The first-order condition is

$$\mu - 1 - \gamma\sigma^2\alpha^* \begin{cases} \leq 0 & \text{if } \alpha^* < w, \\ \geq 0 & \text{if } \alpha^* > w. \end{cases} \quad (16)$$

Note that evaluating (16) at $\alpha^* = 0$ we obtain that $\mu - 1 > 0$, since the mean return on the risky asset is assumed to be greater than one. Thus, $\alpha^* = 0$ cannot be a solution to the asset allocation problem because it violates the first-order condition.

On the other hand, note that $\mu \geq 1 + \gamma\sigma^2w$, we will have that the optimal solution is $\alpha^* = w$. Intuitively, the condition says that if the return on the risky asset surpasses the return on the

safe asset plus a quantity proportional to the wealth of the individual, then the optimal portfolio allocation choice is to invest everything on the risky asset.

Finally, at the interior solution, $0 < \alpha^* < w$ we have that $\mu - 1 - \gamma\sigma^2\alpha = 0$ or $\alpha^* = \frac{\mu-1}{\gamma\sigma^2}$.

In summary, the optimal portfolio $(\alpha^*, \beta^*) : \mathbb{R}^4 \rightarrow \mathbb{R}^2$ is given by the vector function:

$$(\alpha^*, \beta^*)(\mu, \sigma, \gamma, w) = \begin{cases} \left(\frac{\mu-1}{\gamma\sigma^2}, w - \frac{\mu-1}{\gamma\sigma^2} \right) & \text{if } \mu < 1 + \gamma\sigma^2 w \\ (w, 0) & \text{if } \mu \geq 1 + \gamma\sigma^2 w \end{cases} \quad (17)$$

- (b) We have that $u'(x) = \gamma \exp(-\gamma x)$ and $u''(x) = -\gamma^2 \exp(-\gamma x)$. Thus, $r_A(x, u) = \gamma$. Therefore, the parameter γ in the utility function is the coefficient of absolute risk aversion, which implies that the utility function $u(x) = -\exp(-\gamma x)$ has *Constant Absolute Risk Aversion*, otherwise known as a CARA utility function. This type of utility function is used widely in financial economics and finance.
- (c) Given the characterization of the optimal portfolio given in (17), it is easy to see that $\alpha_1^* < \alpha_2^*$, since $\gamma_1 > \gamma_2$ implies that individual 1 is more risk averse than individual 2.

5. (2) $\Rightarrow$ (3). We have that

$$\begin{aligned} u_1(c(F, u_1)) &= \int u_1(z) dF(z) \\ &= \int \psi(u_2(z)) dF(z) \leq \psi \left(\int u_2(z) dF(z) \right) = \psi(u_2(c(F, u_2))) = u_1(c(F, u_2)) \end{aligned}$$

where the first equality is by definition of the certainty equivalent; the second equality is because, by assumption, there exist strictly concave increasing function ψ such that $u_1 = \psi \circ u_2$, the inequality is due to Jensen's inequality; the next-to last equality is by definition of the certainty equivalent; and the last equality again uses the fact that $u_1 = \psi \circ u_2$. Therefore, we have that $u_1(c(F, u_1)) \leq u_1(c(F, u_2))$ and since u_1 is increasing we have that $c(F, u_1) \leq c(F, u_2)$.

(3) $\Rightarrow$ (4). We have that $u_1(c(F, u_1)) = \int u_1(z) dF(z) \geq u_1(c)$, and by monotonicity for money, this implies that $c(F, u_1) \geq c$. By assumption, we also have that $c(F, u_2) \geq c(F, u_1) \geq c$. Since u_2 is increasing we have that $u_2(c(F, u_2)) \geq u_2(c)$ which in turn implies that $\int u_2(z) dF(z) \geq u_2(c)$.

(4) $\Rightarrow$ (2). Define $\psi(x) = u_1(u_2^{-1}(x))$. We will show that ψ is a concave function. Consider three numbers $u_2(x) < u_2(y) < u_2(z)$ in the range of u_2 and let $\lambda \in (0, 1)$ satisfy $u_2(y) = \lambda u_2(x) + (1 - \lambda)u_2(z)$. We will prove that $u_1(y) \geq \lambda u_1(x) + (1 - \lambda)u_1(z)$.

If $u_1(y) < \lambda u_1(x) + (1 - \lambda)u_1(z)$, then for some $\mu > \lambda$ we have $u_1(y) < \mu u_1(x) + (1 - \mu)u_1(z)$ but $u_2(y) > \mu u_2(x) + (1 - \mu)u_2(z)$, that is, $[y] \prec_1 \mu[x] \oplus (1 - \mu)[z]$ and $[y] \succ_2 \mu[x] \oplus (1 - \mu)[z]$, which contradicts (4). Therefore, $[y] \succeq_1 \lambda[x] \oplus (1 - \lambda)[z]$ and $u_1(y) \geq \lambda u_1(x) + (1 - \lambda)u_1(z)$. But given the definition of ψ , the previous inequality implies that $\psi(u_2(y)) \geq \lambda \psi(u_2(x)) + (1 - \lambda)\psi(u_2(z))$, but this is equivalent to ψ being a concave function.

6. (2) $\Rightarrow$ (1). We can write

$$\int u(y) dG(y) = \int \mathbb{E}[u(x + \epsilon)|x] dF(x) \leq \int u(\mathbb{E}[x + \epsilon|x]) dF(x) = \int u(x) dF(x),$$

where the first equality is by the law of iterated expectations and using the assumption that $y = x + \epsilon$, the inequality is Jensen's inequality, since u is concave, and the last inequality uses the fact that $\mathbb{E}[\epsilon|x] = 0$.

(1) $\Leftrightarrow$ (3). First we need an auxiliary result. From the proof of proposition 4.14, we know that for any lottery $F \in \mathcal{F}$, its expected value can be calculated as:

$$\mathbb{E}[F] = \int_w^b z dF(z) = \int_2^b \bar{F}(x) dx + w = \int_w^b (1 - F(x)) dx + w,$$

where we have used $u(z) = z$. Therefore, we have that

$$\int_w^b F(x)dx = b - \mathbf{E}[F]. \quad (18)$$

Now, to show that (1) is equivalent to (3), define the mapping $\xi(t) : \mathbb{R} \rightarrow \mathbb{R}$ by

$$\xi(t) = \int_w^t [F(x) - G(x)]dx.$$

Clearly, $\xi(w) = 0$ and since F and G have the same mean, using (18), it is also true that $\xi(b) = 0$. Now, we have that

$$\begin{aligned} \int_w^b u(x)dF(x) - \int_w^b u(x)dG(x) &= \int_w^b u(x)d(F(x) - G(x)) \\ &= u(x)[F(x) - G(x)]_w^b - \int_w^b (F(x) - G(x))u'(x)dx \\ &= - \int_w^b (F(x) - G(x))u'(x)dx \\ &= - \left\{ u'(x)\xi(x) \Big|_w^b - \int_w^b \xi(x)u''(x)dx \right\} \\ &= \int_w^b \xi(x)u''(x)dx, \end{aligned}$$

where the second inequality is by integration by parts; the third inequality is because $F(x) - G(x)$ have the same value at the endpoints; the fourth equality is by integration by parts; and the last equality is because $\xi(a) = \xi(b) = 0$ which we proved above.

Thus, condition (1) is true if and only if the left hand side is nonnegative for all u with $u''(x) \leq 0$ everywhere. By the equality, the latter holds if and only if $\xi(x) \leq 0$ everywhere, i.e., Condition (3) is true.

- (3) $\Rightarrow$ (2). This part of the problem is by far the hardest one. The general case with arbitrary distribution functions takes quite a bit more math to do it in general. The case of finitely supported distributions is doable, but extremely tedious to write rigorously. However, I can provide some intuition and a proof for a particular case. What we are trying to show here is that the “integral of the c.d.f.s” condition implies the existence of a random variable ϵ , such that if Y is the random variable with distribution G and X is the random variable with distribution F , then $Y = X + \epsilon$; and ϵ is such that it has zero-conditional mean given X , i.e., $\mathbf{E}[\epsilon|X] = 0$. Thus, the crucial object of analysis is the function

$$\zeta_F(t) = \int_w^t F(x)dx, \quad \text{for } t \geq w.$$

For any $F \in \mathfrak{F}$, the function ζ is convex. Moreover, for finitely supported distributions, it is piece-wise linear also. If you are desperate to see how this function looks like, you can go ahead and take a peek at Figure 1, which plots two examples; I will describe this figure in detail below. The function ζ has the property that it is zero until w ; after the last element b of the support of the lottery, it has slope 1. I want to point two facts

- (a) Any function of this sort—that is, piecewise-linear and convex, with value (and slope) zero below some value and with terminal slope 1 after some value— can be “unwound” into a corresponding finitely supported lottery: The location of the kinks constitute the support of the lottery, and the size of the change in slope at the king gives the probability of that value.
- (b) If ζ_F and ζ_G are two of these functions, the “higher” of the two once both hit slope 1 corresponds to the lottery with the greater mean. Therefore, as we have seen previously in (18), F and G have the same mean if and only if ζ_F and ζ_G coincide after the last value in the union of their supports.

Suppose that $\zeta_F(t) \leq \zeta_G(t)$ for every $t \geq w$, and suppose that F and G have the same mean. Then, the heart of the argument is to “convert” ζ_F into ζ_G one step at a time. Each step involves adjusting each piecewise linear segment at a time, moving from right to the left. This is not technically complicated, but very tedious to write rigorously for the lotteries with finite support but with an arbitrary number of points. Therefore, I will focus on lotteries which are finitely supported on just two points. So let X be the random variable representing the lottery $p[x_1] \oplus (1-p)[x_2]$ and let Y be the random variable representing the lottery $q[y_1] \oplus (1-q)[y_2]$, where $x_1 \leq x_2$ and $y_1 \leq y_2$. Obviously, the CDF’s of both lotteries are step functions and are given by

$$F(x) = \begin{cases} 0 & x \in (-\infty, x_1) \\ p & x \in [x_1, x_2) \\ 1 & x \in [x_2, \infty) \end{cases} \quad G(x) = \begin{cases} 0 & x \in (-\infty, y_1) \\ q & x \in [y_1, y_2) \\ 1 & x \in [y_2, \infty) \end{cases}$$

where F is the distribution of X and G is the distribution of Y . The respective “integral of the CDF” functions are given respectively by

$$\zeta_F(t) = \begin{cases} 0 & t \in (-\infty, x_1) \\ p(t - x_1) & t \in [x_1, x_2) \\ t - \mathbf{E}[F] & t \in [x_2, \infty) \end{cases} \quad \zeta_G(t) = \begin{cases} 0 & t \in (-\infty, y_1) \\ q(t - y_1) & t \in [y_1, y_2) \\ t - \mathbf{E}[G] & t \in [y_2, \infty) \end{cases}$$

Now, if $\zeta_F(t) \leq \zeta_G(t)$ for all t , we must have that $y_1 \leq x_1$ and $x_2 \leq y_2$, this is what it is shown in figure 1. Note that both functions start at zero and then after y_2 they both have the same slope of 1. Moreover, since they have the same mean, they coincide with each other after the cutoff y_2 .

In order to transform $\zeta_F(t)$ into ζ_G , what we need to do is shift probability from x_2 to y_1 and y_2 and do the same for x_1 in such a way that, conditional on x_1 and x_2 , the mean of this “shift” is zero. Figure 2 gives an overview of the procedure. This amounts to supplement x_2 with a lottery that, on the upside gives the amount $y_2 - x_2$ and gives the amount $-(x_2 - y_1)$ on the downside; this lottery must have probability λ on the upside and $1 - \lambda$ on the downside such that the conditional mean given x_2 is zero:

$$\lambda(y_2 - x_2) + (1 - \lambda)(-(x_2 - y_1)) = 0 \Rightarrow \lambda = \frac{x_2 - y_1}{y_2 - y_1}.$$

Similarly, we supplement x_1 with a lottery that, on the upside gives $y_2 - x_1$ and gives amount $-(x_1 - y_1)$ on the downside; this lottery must have probability μ on the upside and $1 - \mu$ on the downside such that the conditional mean given x_1 is zero:

$$\mu(y_2 - x_1) + (1 - \mu)(-(x_1 - y_1)) = 0 \Rightarrow \mu = \frac{x_1 - y_1}{y_2 - y_1}.$$

The previous construction visualized using probability trees is illustrated in figure 3. The last tree represents the lottery Y with distribution function G . It can be shown, with some tedious algebra that $q = p(1 - \mu) + (1 - p)(1 - \lambda)$. Notice that we have constructed a random variable ϵ which indeed has conditional mean of zero given X . The joint distribution of (X, ϵ) is given by the following table:

X/ϵ	$y_2 - x_1$	$-(x_1 - y_1)$	$(y_2 - x_2)$	$-(x_2 - y_1)$
x_1	$p\mu$	$p(1 - \mu)$	0	0
x_2	0	0	$(1 - p)\lambda$	$(1 - p)(1 - \lambda)$

You can check that indeed it is the case that this $\mathbf{E}[\epsilon|X] = 0$ and that $Y = X + \epsilon$. This concludes the proof for this particular case. However, note that this is far from what we were initially supposed to prove since this was just a particular case of lotteries with finite support. The general case for finitely supported lotteries with arbitrary number of points is proven in a similar way, via

a constructive proof. The construction is straightforward, but very tedious to write rigorously. On the other hand, proving this for the general case is a bit hard and it requires some technical background to do it rigorously. If you want to see how it is done, see Theorem 2, page 109, in Machina, M., & Pratt, J. (1997). *Increasing risk: some direct constructions*. Journal of Risk and Uncertainty, 14(2), 103-127.

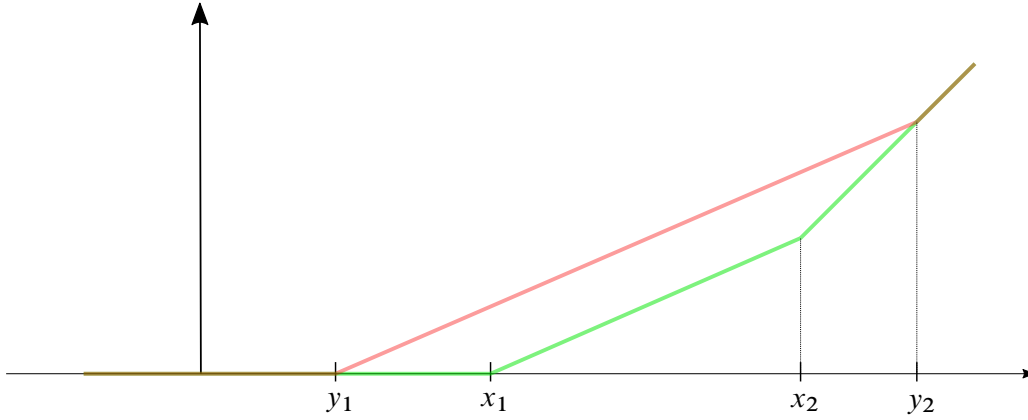


Figure 1: ζ_F in green and ζ_G in red.

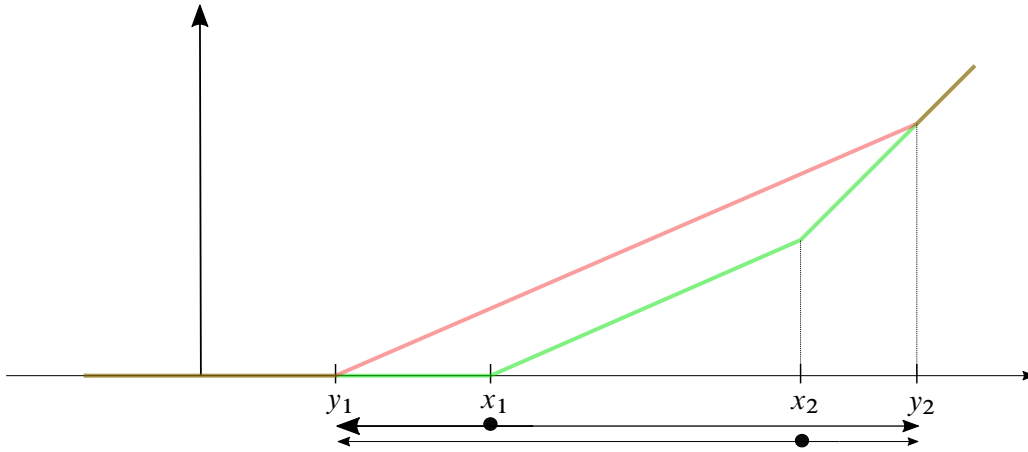


Figure 2: Illustration of the procedure to transform ζ_F into ζ_G .

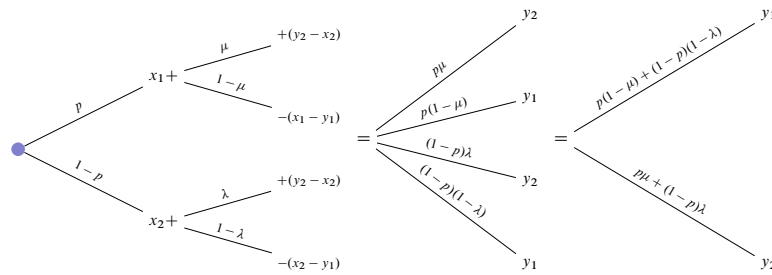


Figure 3: Construction of the conditional lotteries and the probability shifts.

7. Reflexivity Let $F \in \mathfrak{F}$. Then since $F(x) \leq F(x)$ for all x , we have that $F \triangleright_{\text{fosd}} F$.

Antisymmetry Let $F, G \in \mathfrak{F}$. Suppose that $F \triangleright_{\text{fosd}} G$ and that $F \neq G$. This implies that $F(x) \leq G(x)$ for every x and there exists at least one $\hat{x}$ such that $F(\hat{x}) < G(\hat{x})$. Therefore, it is impossible to have $F(x) \geq G(x)$ which implies that $\neg(G \triangleright_{\text{fosd}} F)$.

Transitivity Let $F, G, H \in \mathfrak{F}$ such that $F \triangleright_{\text{fosd}} G$ and $G \triangleright_{\text{fosd}} H$. This implies that $F(x) \leq G(x) \leq H(x)$ for all x . Transitivity of $\leq$ implies that $F(x) \leq H(x)$. Thus, $F \triangleright_{\text{fosd}} H$.